

Internship report - Faisal Haruna Lawan

Faisal joined Intel Programmable Solutions Group (PSG), in the Cloud and Enterprise Acceleration Division (CEAD) UK Engineering team as an intern in July 2023 for a duration of 13 months. Subsequently, Intel PSG has become Altera and all future company references in this document will be to Altera.

During his first month, Faisal completed all required training and orientation activities. He then developed a simple FPGA design which implemented counter logic. Faisal effectively completed this task, demonstrating understanding of design architecture, RTL implementation, the use of simulation tools to verify and debug a design, the use of the Quartus® FPGA Design Tool Suite to configure appropriate timing requirements and compile FPGA images, and installing the design on a development kit to validate that it works as expected on hardware.

Faisal then implemented a search algorithm to detect duplicate Media Access Control (MAC) addresses in Virtio Network devices attached to an FPGA-based system which offloaded network switching functionality to hardware. He analyzed the current algorithmic approach and then explored alternative methodologies to improve efficiency. He performed and reported benchmarking of resource use and execution time for each potential solution before making a recommendation to the software development team.

Faisal continued working on software systems by refactoring a legacy software tool used to manage the Altera Infrastructure Processing Unit. He investigated the existing source code and, together with his technical mentor, identified a work plan to improve code readability, efficiency, and modularity by limiting the tool scope to service Altera's Virtio DMA IP only. Faisal extracted the required functionality from a much larger source code repository and created a standalone software tool according to the project specification. He also developed a python-based testing framework to validate his changes against the required functionality.

Faisal spent the second half of his internship on an implementation of Google's Snappy Compression algorithm in an FPGA. He began with detailed research into Google's Snappy Compression and Decompression algorithms. His investigation included review of the publicly available code, writing his own software implementation of the algorithm to reinforce his understanding, suitable testing and refinement of his implementation in software to achieve nearly identical results to the Google code, and presentation of this material to key stakeholders within Altera's Engineering organization. This presentation was of excellent quality and provided meaningful learnings to those attending.

He then focused on the implementation of the Compression algorithm. He undertook the architecture and design of the implementation with minimal assistance from other staff, working efficiently and independently. This included researching methods to perform the dictionary construction, RTL module design in SystemVerilog, module-level simulation of

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his blocks to ensure correct behavior, learning the function and usage of existing IP to fulfill certain functions within his design, and acceptance of feedback from staff while providing data-driven reasoning for his design decisions.

Once his module was completed and verified in simulation, he then developed a NIOS® V based test system to implement this design on an FPGA development board. He exercised strong interpersonal skills and engineering collaboration with his intern peers to develop this system, leveraging others' expertise to enhance his own understanding. He developed software to drive the embedded processor system, which allowed test files to be compressed and then verified using the Google Snappy Decompression algorithm he developed. This design was demonstrated to the CEAD engineering team, as well as other global stakeholders.

Faisal continued to analyze the design, suggesting, researching, implementing, and testing improvements to the hardware implementation prior to his departure.

Throughout his internship, Faisal has participated in various company-organized volunteer activities, ranging from student outreach to community housing renovation.

Faisal has worked with considerable independence and self-motivation throughout his placement. He has demonstrated behaviors, skills, and quality of work equivalent to or better than a junior engineer with some years' experience. I strongly recommend considering Faisal for any role to which he applies as I believe he will bring considerable value to any organization.

Frederic Richard (Manager)
Engineering Manager

Andrew Bown (Mentor)
SoC Design Engineer